



Amorphous silicon-lithium niobate thin film strip-loaded waveguides

YIWEN WANG, ZHIHUA CHEN, LUTONG CAI, YUNPENG JIANG, HOUBIN ZHU, AND HUI HU*

School of Physics, Shandong University, Jinan, 250100, China

**hhu@sdu.edu.cn*

Abstract: The heterogeneous integration of an amorphous silicon (a-Si) film with a lithium niobate (LN) thin film combines both the mature micro-processing technology of Si and the excellent optical properties of LN. An a-Si thin film was deposited on an LN thin film, and strip-loaded waveguides were designed, fabricated, and characterized. A full-vectorial finite difference method was used to explore the single-mode conditions and appropriate dimensions for the strip-loaded waveguides. The waveguide mode size could be as small as $0.36\ \mu\text{m}^2$. By adjusting the thickness and width of the a-Si loading strip, the distribution of light power could be mainly confined in the LN layer. The maximal light power that could be confined in LN was 91%, which was obtained at an a-Si thickness of 65 nm. A set of waveguides with widths of 2–7 μm were prepared by inductively coupled plasma (ICP) etching of the a-Si thin film. Following annealing at 300°C in air for 1 hour, light transmission was observed in the waveguide. The 2- μm -wide waveguide showed propagation losses of 20 dB/cm for the quasi-TM (q-TM) mode and 42 dB/cm for the quasi-TE (q-TE) mode at 1550 nm. The root-mean-square (RMS) surface roughness of the a-Si thin film before and after annealing was 1.04 and 0.35 nm, respectively. High-resolution transmission electron microscopy (HRTEM) was performed to investigate the interface morphologies. A well-defined interface was clearly observed, and the structure of the a-Si thin film was proved to be amorphous.

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1. Introduction

LN is an attractive multifunctional material exhibiting many advantages, such as excellent electro-optic, nonlinear optic and acousto-optic properties and good transmittance at both visible and telecommunication wavelengths [1,2]. Recently, single-crystal LN thin films on a low-refractive-index SiO₂ cladding layer (LN-on-insulator, LNOI) were prepared using ion implantation and wafer bonding technologies [3–5]. Due to the large refractive index difference of LN and SiO₂, good optical confinement and strong light guiding are realized, and accordingly, various high-performance photonic devices, including electro-optical modulators [6–8], photonic crystals [9,10], wavelength conversion devices [11,12] and domain inversion on LNOI [13,14], have been fabricated. A waveguide is a basic structure for photonic devices. Several types of waveguide fabrication technologies on LNOI have been introduced, such as etching [15], proton exchange [16], and strip-loaded structures [17]. In strip-loaded waveguides, a silicon nitride loading strip on an LN thin film (SiN-LNOI) has been successfully demonstrated for use in electro-optical modulators [18], second-harmonic generation [19], and integrated platforms [20]. Similar to SiN, Si is another important strip-loading material. Si possesses unparalleled advantages in processing technology and fabrication cost [21–23]. Due to the high refractive index of Si, the bending radius of Si-LNOI waveguides can be as small as 15 μm , which gives a bend loss below 10^{-7} , according to the simulation. Furthermore, Si layers have been extensively used in integrated optics [24,25].

A Si-bulk-LN platform (Si-LN) has been developed to exploit the linear electro-optical effect in the mid-IR range [26]. A compact modulator has also been realized via the deposition of a-Si onto an x-cut LN substrate [27]. Compared to the Si-LN platform, the combination of Si and a single-crystal LN thin film (Si-LNOI) could significantly reduce the waveguide mode cross-section, resulting in optical devices with small volumes and high efficiencies.

The combination of an LN thin film and silicon-on-insulator (SOI) technology has been attempted. LN thin film/SOI electro-optic tunable ring resonators have been demonstrated [28], and a compact modulator based on an LN thin film bonded to SOI ring resonators has been presented [29]. Integrated optical building blocks have been successfully achieved by bonding LNOI to SOI (LNOI-SOI) [30]. In the above material structure, the LN thin film was coated on the Si thin film. It was necessary to structure the Si thin film prior to its bonding with the LN thin film, making the direct bonding process challenging. Furthermore, electrode fabrication and connection would be another challenging step for this LNOI-SOI combination. If an Si thin film could be coated on an LN thin film, due to the abundance and facile fabrication of Si, the combination of an Si thin film and LNOI (Si-LNOI) could provide great design and fabrication flexibility. The deposition of an Si thin film onto LNOI is a simple approach for the realization of an Si-LNOI structure. Plasma-enhanced chemical vapor

deposition (PECVD)-deposited a-Si has been widely used in integrated optics [31]. A-Si rib waveguides on an Si wafer have been fabricated, and a propagation loss as low as 1 dB/cm has been reported [32]. Thus, a-Si could be a potential loading strip material for waveguides, and a-Si directly deposited onto LNOI by PECVD could be a promising platform for integrated optics.

In this work, an a-Si thin film was deposited onto LNOI by PECVD. Si strip-loaded waveguides were fabricated and investigated. The near-field optical intensity distribution, single-mode condition, optical power distribution and mode size in the strip-loaded waveguides were investigated using a full-vectorial finite difference method. The simulated minimum mode size could be as small as $0.36 \mu\text{m}^2$. The light power maximum confined in the LN layer obtained via simulation was 91%. Si-LNOI waveguides with widths of 2–7 μm were prepared, and light transmission was observed in the waveguides. In a 2- μm -wide waveguide, propagation losses of 20 dB/cm and 42 dB/cm were obtained for the q-TM and q-TE modes, respectively. The top-view and cross-section of the waveguides were examined by scanning electron microscopy (SEM). The surface topography of the a-Si thin films before and after annealing were measured by atomic force microscopy (AFM), and the RMS surface roughness was 1.04 and 0.35 nm, respectively. Clear interfaces were observed in the HRTEM measurement.

2. Simulation

Figure 1(a) shows the cross-section of an Si-LNOI waveguide. The components from the bottom to the top are the LN substrate, SiO_2 cladding layer, LN layer and a-Si loading strip.

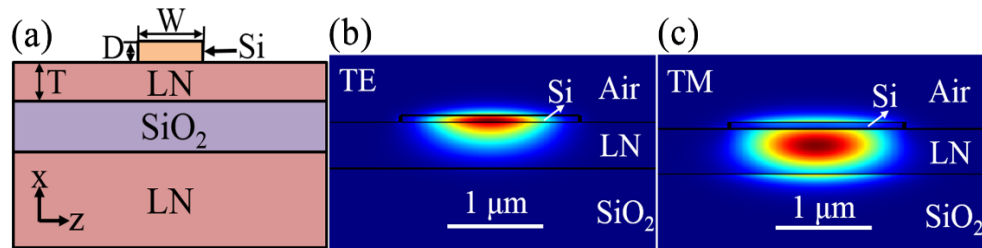


Fig. 1. (a) Schematic cross-section of the Si-LNOI waveguide. Simulated electric field intensity distributions of the fundamental (b) q-TE mode and (c) q-TM mode in a 2- μm -wide waveguide at 1.55 μm .

To investigate the mode distribution of the waveguide, a full-vectorial finite difference method calculation was performed using the commercial Mode Solutions software package produced by Lumerical. The LNOI were all x-cut, and the birefringence of the LN material was considered in the simulations. The extraordinary and ordinary refractive indices of the LN thin film, as measured by a prism coupler (Model 2010 Metricon) at 1539 nm, were 2.1383 and 2.2120, respectively. The refractive index of a-Si, as measured by an ellipsometer, was 3.16 at 1550 nm. The refractive index of deposited Si is related to the deposition parameters, such as SiH_4/He flow rate, RF power, pressure and temperature. A wide range of deposited Si refractive indices, from 2.59 to 3.73, has been reported [33–35]. Our value is consistent with those in the literature [35]. Both q-TE and q-TM modes were supported in the strip-loaded waveguide. Figures 1(b) and 1(c) show simulated electric field intensity profiles (at a wavelength of 1550 nm) for the q-TE and q-TM modes for an a-Si strip with a thickness and width of 70 nm and 2 μm , respectively. The mode distribution difference between the q-TE and q-TM modes mainly lies in the geometry of the waveguide, such as the thicknesses and the refractive indices of the Si and LN thin films [36].

To investigate the relation between the waveguide geometry and the single-mode conditions, the parameters of the waveguides (the thickness T of the LN layer and the width W and thickness D of the a-Si strip) were varied in the simulation. The value of T was

selected to ensure that only one electric field intensity maximum was supported in the vertical direction of the LN thin film. For this LN planar waveguide, the single-mode condition, $0 < T < 0.59 \mu\text{m}$ for the q-TE mode and $0.09 \mu\text{m} < T < 0.57 \mu\text{m}$ for the q-TM mode, was identified. For the Si-LNOI waveguides, the single-mode condition was determined by the values of T , W and D . Figure 2 shows the single-mode condition for a fixed LN layer thickness of $T = 0.5 \mu\text{m}$. The curves show the boundary between the single-mode and multi-mode conditions. As the a-Si loading strip thickness D increases, the a-Si width W should decrease in order to satisfy the single-mode condition. For example, for a 70-nm-thick a-Si loading strip, the value of W should be between 0 and $1.22 \mu\text{m}$ for the q-TE mode and between 0 and $1 \mu\text{m}$ for the q-TM mode for a waveguide performing in a single mode. The cut-off width of the loading strip may be close to zero micrometers [37].

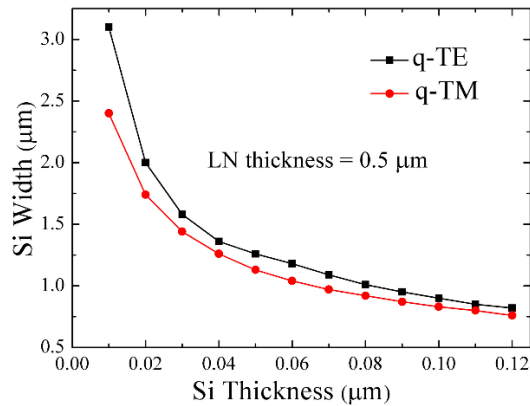


Fig. 2. Single-mode conditions for the Si-LNOI waveguides at a wavelength of 1550 nm.

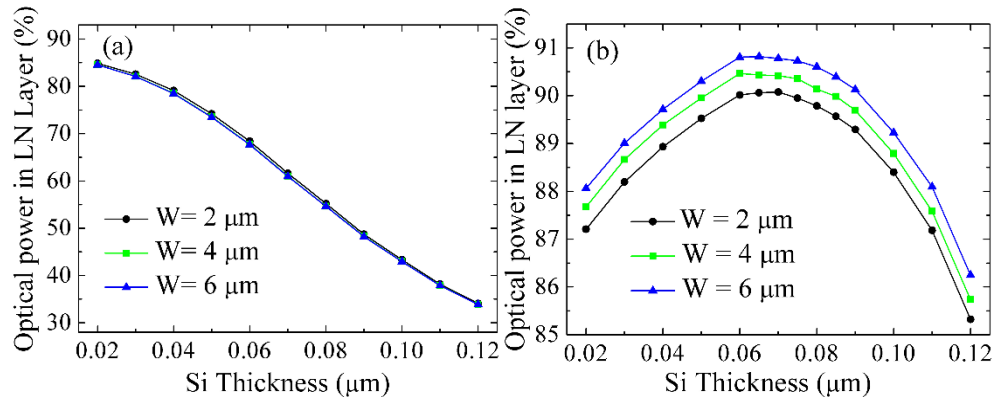


Fig. 3. Relationship between the optical power distribution in a 500-nm LN layer and the thickness (T) and width (W) of an a-Si loading strip for the (a) q-TE mode and (b) q-TM mode.

The primary optical power distribution in the LN layer and the a-Si loading strip is another important factor for the waveguide performance. Most of the optical power was expected to be confined in the LN layer based on its excellent electro-optical and nonlinear optical properties. The optical power in the LN layer as a function of W is displayed in Fig. 3. As shown in Fig. 3(a), the optical power of the q-TE mode in the LN layer decreased as the a-Si thickness increased. The a-Si width had little influence on the optical power distribution. As shown in Fig. 3(b), the optical power of the q-TM mode in the LN layer initially increased as the a-Si thickness increased. A maximum value was obtained at an a-Si thickness of approximately $0.065 \mu\text{m}$. As the a-Si thickness increased further, the optical power in the LN

layer was reduced. Thus, the width of the a-Si layer had a strong influence on the optical power distribution for the q-TM mode.

To strengthen nonlinear effects such as the difference frequency generation [38] and electro-optical phase modulation [39], a small mode size is preferred. Based on the previous simulation results, the thickness values of the LN layer and a-Si loading strip were selected to be $0.5\ \mu\text{m}$ and $0.065\ \mu\text{m}$, respectively. The simulation results for the relationship between the mode size and the width W of the a-Si loading strip are shown in Fig. 4. The minimum mode size was $0.41\ \mu\text{m}^2$ (product of the positions of the $1/e$ light intensity in the horizontal and vertical directions) at $W = 1\ \mu\text{m}$ for the q-TE mode. The minimum mode size was $0.36\ \mu\text{m}^2$ at $W = 0.6\ \mu\text{m}$ for the q-TM mode. As the width of the a-Si loading strip increased from that for the minimum mode size, both the q-TE and q-TM sizes became larger. When the width of the a-Si loading strip decreased from that for the minimum mode size, the light confinement became weak and a larger mode size was observed.

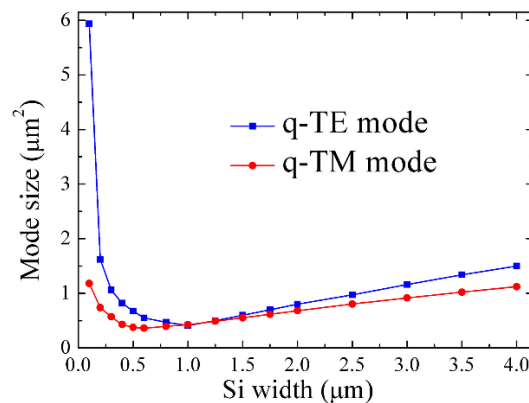


Fig. 4. Relationship between the calculated mode size and the width of an a-Si loading strip for the q-TE mode and q-TM mode.

3. Experiments and results

X-cut LNOI samples (acquired from NANOLN) were prepared by ion implantation and wafer bonding technologies. One sample was used to measure the refractive index of the a-Si thin film, whereas other samples were used to fabricate the Si-LNOI waveguides. The fabrication process is shown schematically in Fig. 5. A 70-nm-thick a-Si thin film was deposited on the surface of the LN film layer via PECVD. The deposition parameters of the a-Si film are given in Table 1. The thickness of the a-Si film was measured by a step profiler, and the Si film was confirmed to be amorphous by Raman scattering measurements. A Raman peak was observed at $480\ \text{cm}^{-1}$, consistent with that of the a-Si [40]. A photoresist mask with 2 to 7- μm -wide channels (waveguide width) along the y-axis of the LN film was patterned via photolithography. The a-Si loading strips were etched using ICP. The remaining photoresist was removed by acetone. The end faces of the waveguide were polished using the chemical mechanical polishing (CMP) method in order to facilitate end-face coupling in the optical characterization. The length of the waveguide was 1 mm. Figures 6(a) - 6(c) show the optical microscopy images of the Si-LNOI waveguides. Figure 6(a) shows the top view of the Si-LNOI waveguides with a polished end face. Figure 6(b) shows the enlarged view of a 2- μm -wide waveguide. The cross-section of the polished end face is presented in Fig. 6(c). From top to bottom, the LN thin film (the bright stripe), SiO_2 cladding (the dark stripe), and LN substrate are observed. Figures 6(d) and 6(e) show the top-view and cross-section SEM images of the Si-LNOI waveguide. In Fig. 6(e), the cross-section of the Si strip, LN thin film, SiO_2 layer, and LN substrate are observed. These figures show the sharp polished end face produced by the CMP process.

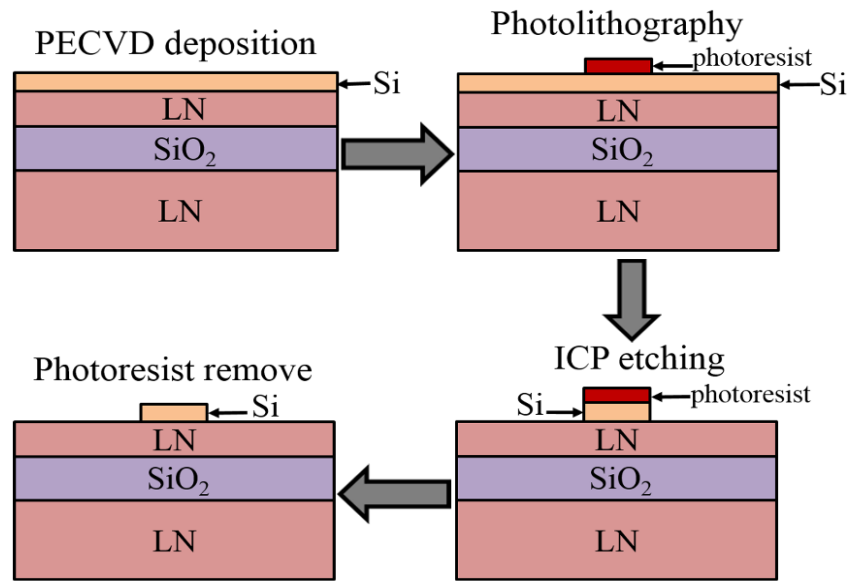


Fig. 5. Fabrication process for an Si-LNOI waveguide.

Table 1. Si deposition parameters

SiH ₄ flow (sccm)	He flow (sccm)	RF Power (W)	Substrate temperature (°C)	Pressure (mTorr)
45	450	40	350	1500

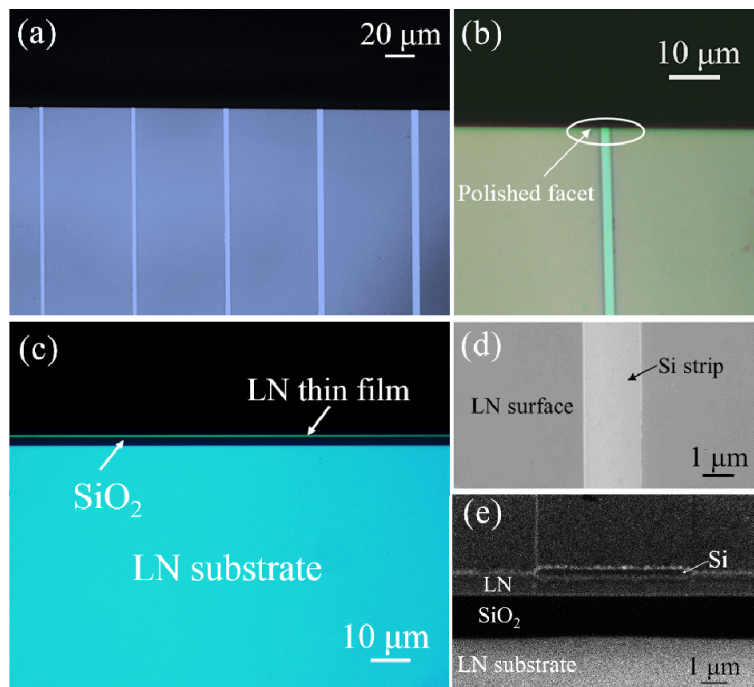


Fig. 6. Optical microscopy images of (a) the top view, (b) the enlarged view of a 2-μm-wide a-Si loading strip with a polished end face and (c) the cross-section of the sample. SEM images of (d) the top view of a 2-μm-wide a-Si loading strip and (e) the cross-section of a 7-μm-wide a-Si strip-loading waveguide. The thicknesses of the a-Si layer, the LN thin film and the SiO₂ layer were 70 nm, 500 nm and 2 μm, respectively.

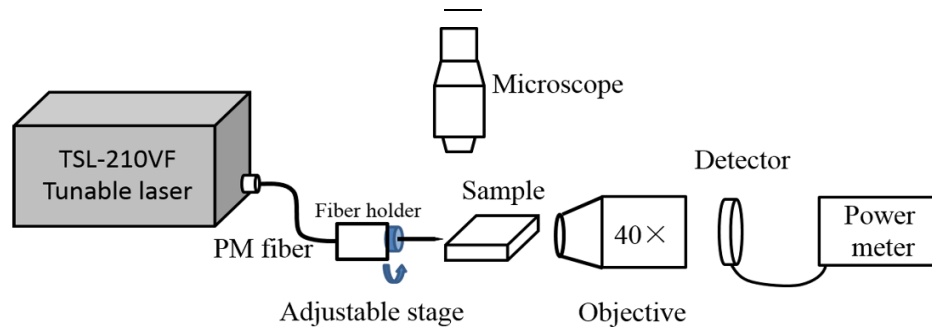


Fig. 7. Schematic of the coupling system.

As shown in Fig. 7, a tunable semiconductor laser (Santec TSL-210) was used as the light source in the experiment. Linearly polarized light emitted from the laser was transmitted through a polarization-maintaining (PM) fiber. The output end of PM fiber was fixed on a fiber holder with an adjustable stage. The polarization of the output light was adjusted by rotating the PM fiber. Before the light was coupled to the waveguide through a lensed tip at the end of the fiber, the output of the lensed fiber was carefully adjusted to be TE/TM polarization calibrated by polarizers. A $40\times/0.65$ objective was used to collect the output light. Collimation among the fiber, sample and objective was achieved by adjusting their positions under the optical microscope. A germanium photodiode was then used to measure the output light collected by the objective. The polarization at the waveguide output was checked by a polarizer. When the lensed fiber output was TE polarization, the waveguide output was TE polarization, and there was no TM polarization content, and vice versa for TM polarization. The results showed that there was no polarization rotation in the waveguides. In the loss measurement, there was no polarizer at the output side of the waveguide when the lensed fiber was calibrated to be TE/TM polarization. The position of the input PM fiber was adjusted according to the readings on the power meter. The launch position was determined when the optical power meter reached the maximum value. In this way, the coupling efficiency was optimized. Due to the high optical absorption, light transmission in the waveguide was not observed. This phenomenon may be due to the large density of point defects and dangling bonds generated during the PECVD process [24,41]. Since post-PECVD annealing can reduce the number of point defects and dangling bonds in a-Si [35,42,43], the Si-LNOI samples were annealed in air for 1 hour at 300°C and 350°C , respectively. And the light transmission was observed only for the former. A waveguide between the two polished end faces is considered a Fabry-Perot cavity. Based on the Fabry-Perot resonator method [17,44], the propagation loss α can be evaluated from the following equations:

$$\alpha = \frac{4.34}{L} \left(\ln R - \ln \tilde{R} \right), \text{ where } \tilde{R} = \frac{1}{K} \left(1 - \sqrt{1 - K^2} \right) \text{ and } K = \frac{I_{\max} - I_{\min}}{I_{\max} + I_{\min}} \quad (1)$$

where α was determined by analyzing the contrast K of the cavity resonances and $I_{\max}$ and $I_{\min}$ are the maximum and minimum intensity of the transmitted light, as obtained from the transmission spectrum in Fig. 8. The end-face reflectivity (R) is the fraction of reflected optical power returned to the fundamental mode. In a strong confinement waveguide, R cannot be approximated by the Fresnel result for plane waves of normal incidence at a LN-air interface (11.8% for TE), as is usually done for weakly guiding structures [45]. R has a strong impact on the loss value. Therefore, R must be calculated by the three-dimensional finite difference time domain (FDTD) method [46]. The R values for the $2\text{-}\mu\text{m}$ -wide waveguide were 0.195 for the q-TE mode and 0.134 for the q-TM mode. The waveguide length L was 1 mm in the experiment. The propagation losses were 42 and 20 dB/cm for the q-TE and q-TM modes, respectively. The optical power of the q-TE mode confined in the Si layer was larger

than that for the q-TM mode, and therefore, the propagation loss of the q-TE mode caused by the a-Si loading strip was larger than that of the q-TM mode.

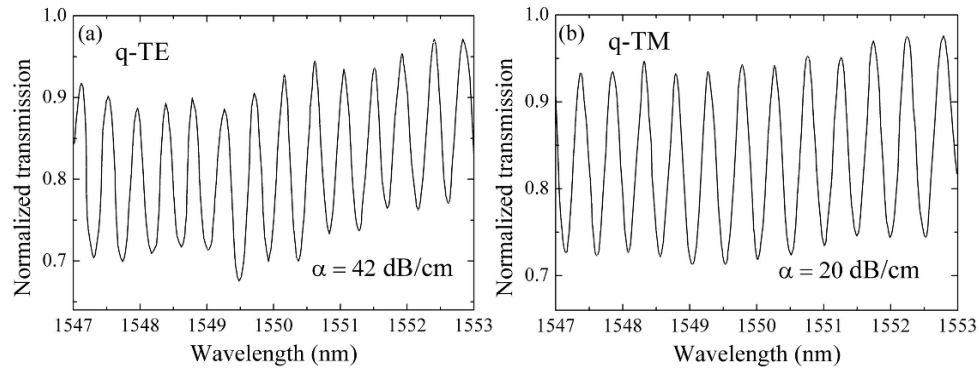


Fig. 8. Normalized transmission of (a) q-TE and (b) q-TM polarized light in a 2- μ m-wide Si-LNOI waveguide as a function of wavelength.

The morphologies of the Si thin film before and after annealing were investigated by AFM, as shown in Fig. 9. The sample was scanned over a $1\ \mu\text{m} \times 1\ \mu\text{m}$ area. The RMS surface roughness of the Si strip before and after annealing was 1.04 and 0.35 nm, respectively. In Fig. 9(a), compact, small-sized, bean-shaped islands were observed. After annealing at 300°C for 1 hour, no obvious islands could be observed, as shown in Fig. 9(b), and the height difference decreased markedly, corresponding to the decreased RMS surface roughness. This result indicated that the surface smoothness was improved after annealing, which might enhance the transmission performance of the Si-LNOI waveguide.

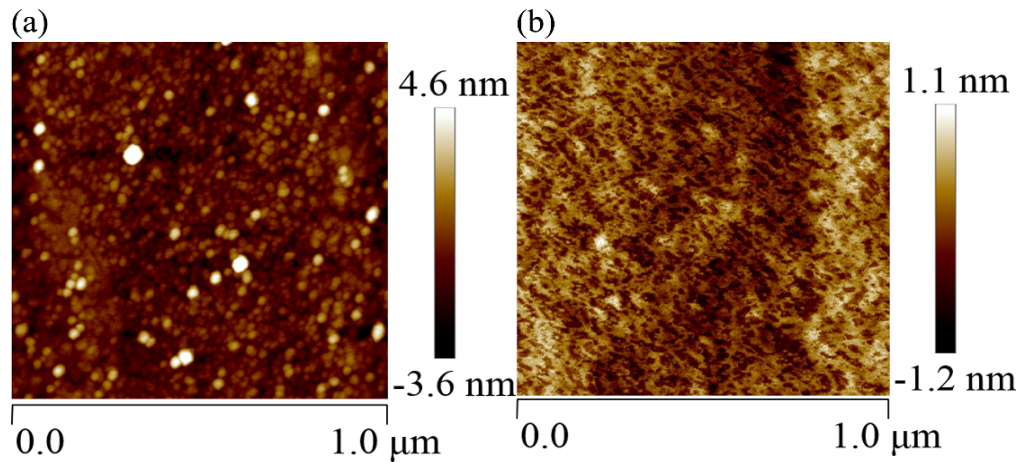


Fig. 9. AFM images of an a-Si loading strip (a) before and (b) after annealing at 300°C for 1 hour.

To investigate the crystalline microstructure of Si before and after annealing, cross-section TEM measurements were employed. Figure 10(a) shows the low-resolution TEM image in which the Si-LNOI structure is clearly seen. The thickness of the a-Si thin film was 70 nm, which agrees with the previous measurements. The HRTEM images of the a-Si film before and after annealing are shown in Figs. 10(b) and 10(c), respectively. The well-defined interface between a-Si and LN can be clearly observed. The typical amorphous structures are present in both images, indicating that the annealing process has a very limited effect on the lattice structure of Si. The interface region between LN and SiO_2 is shown in Fig. 10 (d), and

a clear interface is observed. In addition, the ordered lattice arrangement indicates the good single crystallinity of the LN film.

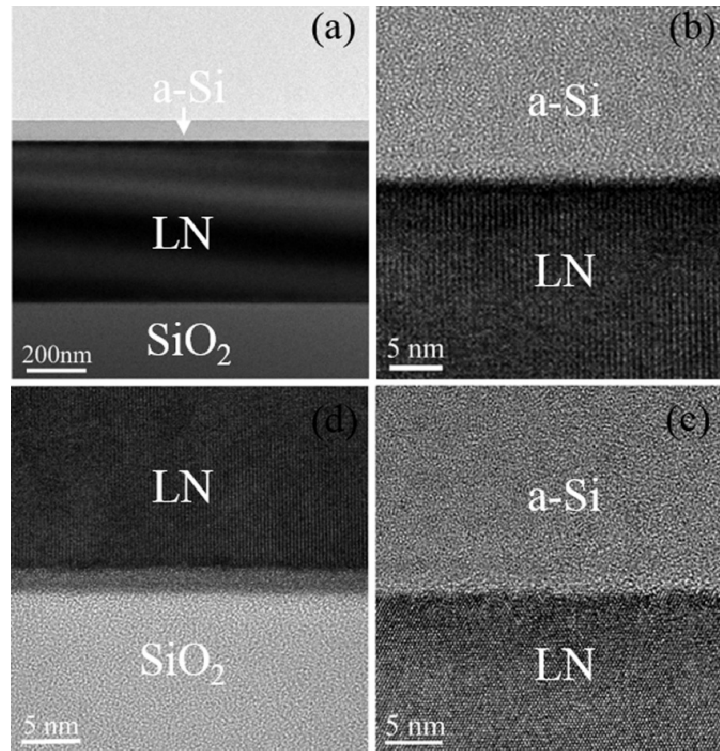


Fig. 10. (a) Low magnification cross-section TEM image of the Si-LNOI sample; HRTEM micrograph of the interface between the a-Si thin film and the LN thin film (b) before and (c) after annealing at 300°C for 1 hour; (d) HRTEM micrograph of the interface between the LN thin film and the SiO₂ cladding layer.

4. Conclusions

In summary, a-Si was deposited onto LNOI, and Si-LNOI waveguides were theoretically and experimentally investigated. The fundamental mode distribution, single-mode condition, optical power distribution and mode size of the strip-loaded waveguide were simulated. The simulated minimal mode size and maximal light power confined in the LN layer was 0.36 μm^2 and 91%, respectively. Si-LNOI waveguides with widths of 2–7 μm were prepared, and light transmission in the waveguide was observed. A 2- μm -wide Si-LNOI waveguide exhibited propagation losses of 20 dB/cm and 42 dB/cm for the q-TM and q-TE modes, respectively. Different deposition methods and parameters were tested to obtain low-loss a-Si. The crystal structures along with the surface and interface morphologies of Si before and after annealing were investigated by AFM and TEM, respectively. The RMS surface roughness before and after annealing was 1.04 and 0.35 nm, respectively. Well-defined interfaces between a-Si and LN and between LN and SiO₂ were clearly observed. Since the optical absorption band of Si is located at 1100 nm, the Si-LNOI waveguide can be used for applications such as modulators and nonlinear optical devices at wavelengths above 1100 nm. This Si-LNOI waveguide realized the promising combination of the excellent optical properties of LN and the mature fabrication technology for Si, laying a foundation for the development of complex, high-performance photonic devices and circuits.

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